

# Abbreviations

This list clarifies the abbreviations and acronyms in Chapters 1–30 and 35, as well as “standard use” acronyms in Chapters 31–34. Acronyms that are specific to the standards considered in Chapters 31 (LTE), 32 (NR), 33 (Wi-Fi), and 34 (Bluetooth and Zigbee), as well as Appendix 30 (GSM, WCDMA, DECT) are given in dedicated glossaries at the end of those chapters.

|        |                                                        |
|--------|--------------------------------------------------------|
| 2G     | Second Generation                                      |
| 3G     | Third Generation                                       |
| 3GPP   | Third Generation Partnership Project                   |
| 3GPP2  | Third Generation Partnership Project 2                 |
| 3SQM   | Single Sided Speech Quality Measure                    |
| A/D    | Analog to Digital                                      |
| AB     | Access Burst                                           |
| ABS    | Almost Blank Subframe                                  |
| AC     | Access Category (Chapter 33)                           |
| AC     | Alternate Current                                      |
| ACCH   | Associated Control CHannel                             |
| ACELP  | Algebraic Code Excited Linear Prediction               |
| ACF    | AutoCorrelation Function                               |
| ACI    | Adjacent Channel Interference                          |
| ACK    | ACKnowledgment                                         |
| ACLR   | Adjacent Channel Leakage Ratio                         |
| ACM    | Address Complete Message                               |
| AD     | Access Domain                                          |
| ADC    | Analog to Digital Converter                            |
| ADDTTS | ADD Traffic Stream                                     |
| ADF    | Average Duration of Fades                              |
| ADPCM  | Adaptive Differential Pulse Code Modulation            |
| ADPM   | Adaptive Differential Pulse Modulation                 |
| ADPS   | Angular Delay Power Spectrum                           |
| ADSL   | Asymmetric Digital Subscriber Line                     |
| AE     | Auto Encoder                                           |
| AF     | Amplify-and-Forward                                    |
| AGC    | Automatic Gain Control                                 |
| AI     | Artificial Intelligence                                |
| AIC    | Akaike Information Criterion                           |
| ALOHA  | a random access packet radio system                    |
| AM     | Amplitude Modulation                                   |
| AMC    | Adaptive Modulation and Coding                         |
| AMPS   | Advanced Mobile Phone System                           |
| AMR    | Adaptive Multi Rate                                    |
| AN     | Access Network                                         |
| ANSI   | American National Standards Institute                  |
| AODV   | Ad hoc On-Demand Distance Vector                       |
| AP     | Access Point                                           |
| APS    | Angular Power Spectrum                                 |
| AR     | Augmented Reality                                      |
| ARIB   | Association of Radio Industries and Businesses (Japan) |
| ARQ    | Automatic Repeat reQuest                               |
| ASIC   | Application Specific Integrated Circuit                |
| ASK    | Amplitude Shift Keying                                 |
| ATIS   | Alliance for Telecommunications Industry Solutions     |
| ATM    | Asynchronous Transfer Mode                             |

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|          |                                                                                               |
|----------|-----------------------------------------------------------------------------------------------|
| ATSC-M/H | Advanced Television Systems Committee – Mobile/Handheld                                       |
| AV       | Audio and Video                                                                               |
| AVC      | Advanced Video Coding                                                                         |
| AWGN     | Additive White Gaussian Noise                                                                 |
| BAM      | Binary Amplitude Modulation                                                                   |
| BAN      | Body Area Network                                                                             |
| BC       | Broadcast (Channel)                                                                           |
| BCH      | Bose–Chaudhuri–Hocquenghem (code)                                                             |
| BCJR     | initials of the authors of Bahl et al. [1974]                                                 |
| BEC      | Backward Error Correction                                                                     |
| BEOL     | Back End Of Line                                                                              |
| BER      | Bit Error Rate                                                                                |
| BFSK     | Binary Frequency Shift Keying                                                                 |
| BICM     | Bit Interleaved Coded Modulation                                                              |
| BLAST    | Bell labs LAYered Space Time                                                                  |
| BLE      | Bluetooth Low Energy                                                                          |
| BN       | Bit Number                                                                                    |
| BOC      | Binary Offset Carrier modulation                                                              |
| BP       | Breakpoint                                                                                    |
| BPF      | BandPass Filter                                                                               |
| BPPM     | Binary Pulse Position Modulation                                                              |
| BPSK     | Binary Phase Shift Keying                                                                     |
| BPTT     | Back Propagation Through Time                                                                 |
| BRN      | Barrage Relay Network                                                                         |
| BS       | Base Station                                                                                  |
| BU       | Bad Urban                                                                                     |
| BW       | Bandwidth                                                                                     |
| C/A      | Coarse Acquisition sequence                                                                   |
| CA       | Cell Allocation                                                                               |
| CAA      | Channel Access Algorithm                                                                      |
| CAF      | Compute and Forward                                                                           |
| CAP      | Controlled Access Period                                                                      |
| CAZAC    | Constant Amplitude Zero AutoCorrelation                                                       |
| CB       | Citizens' Band                                                                                |
| CC       | Country Code                                                                                  |
| CCF      | Cross Correlation Function                                                                    |
| CCI      | Co Channel Interference                                                                       |
| CCITT    | Commitee' Consultatif International de Telegraphique et Telephonique                          |
| CCK      | Complementary Code Keying                                                                     |
| CCSA     | China Communications Standards Association                                                    |
| CD       | Compact Disk                                                                                  |
| cdf      | Cumulative Distribution Function                                                              |
| CDD      | Cyclic Delay Diversity                                                                        |
| CDG      | CDMA Development Group                                                                        |
| CDMA     | Code Division Multiple Access                                                                 |
| CDN      | Content Delivery Network                                                                      |
| CELP     | Code Excited Linear Prediction                                                                |
| CEPT     | European Conference of Postal and Telecommunications Administrations                          |
| CF       | Compress-and-Forward                                                                          |
| CF-Poll  | Contention-Free Poll                                                                          |
| CFO      | Carrier Frequency Offset                                                                      |
| CI       | Close-In channel model (Chapter 7)                                                            |
| CI       | Cell Identify                                                                                 |
| C/I      | Carrier-to-Interference ratio                                                                 |
| CLEAN    | A deconvolution algorithm (written as capital letters, but it is not an acronym for anything) |
| CM       | Connection Management                                                                         |
| CMA      | Characteristic Mode Analysis (Chapter 8)                                                      |
| CMA      | Constant Modulus Algorithm                                                                    |
| CMOS     | Complementary Metal Oxide Semiconductor                                                       |
| CMP      | Cube Map Projection                                                                           |
| CN       | Core Network                                                                                  |

|          |                                                                        |
|----------|------------------------------------------------------------------------|
| CND      | Core Network Domain                                                    |
| CNG      | Comfort Noise Generation                                               |
| CNN      | Convolutional Neural Network                                           |
| CoMP     | Cooperative MultiPoint                                                 |
| CoMP-CB  | Cooperative MultiPoint – Coordinated Beamforming                       |
| CoMP-JP  | Cooperative MultiPoint – Joint Processing                              |
| COST     | European COoperation in the field of Scientific and Technical research |
| COT      | Channel Occupancy Time                                                 |
| CP       | Cyclic Prefix                                                          |
| CPC      | Cognitive Pilot Channels                                               |
| CPFSK    | Continuous Phase Frequency Shift Keying                                |
| CPU      | Central Processing Unit                                                |
| CQI      | Channel Quality Indicator                                              |
| CRA      | Collision/Conflict Resolution Algorithm                                |
| CRC      | Cyclic Redundancy Check                                                |
| CRI      | Collision Resolution Interval                                          |
| CRLB     | Cramer Rao Lower Bound                                                 |
| CS-ACELP | Conjugate Structure-Algebraic Code Excited Linear Prediction           |
| CSD      | Cyclic Shift Diversity                                                 |
| CSI      | Channel State Information                                              |
| CSIR     | Channel State Information at the Receiver                              |
| CSIT     | Channel State Information at the Transmitter                           |
| CSMA     | Carrier Sense Multiple Access                                          |
| CSMA/CA  | Carrier Sense Multiple Access with Collision Avoidance                 |
| CT       | Core network and Terminals                                             |
| CTS      | Clear To Send                                                          |
| CTT      | Cellular Text Telephony                                                |
| CTU      | Coding Tree Unit                                                       |
| CU       | Central Unit                                                           |
| CUG      | Closed User Group                                                      |
| CW       | Continuous Wave                                                        |
| CW       | Contention Window (Chapters 18, 33)                                    |
| D-BLAST  | Diagonal BLAST                                                         |
| DAA      | Detect and Avoid                                                       |
| DAB      | Digital Audio Broadcasting                                             |
| DAC      | Digital to Analog Converter                                            |
| DAF      | Diversity-Amplify-and-Forward                                          |
| DAM      | Diagnostic Acceptability Measure                                       |
| DARPA    | Defense Advanced Research Project Agency                               |
| DASH     | Dynamic Adaptive Streaming over Http                                   |
| dB       | Decibel                                                                |
| DB       | Dummy Burst                                                            |
| DBPSK    | Differential Binary Phase Shift Keying                                 |
| DC       | Direct Current                                                         |
| DCM      | Directional Channel Model                                              |
| DCS1800  | Digital Cellular System at the 1800-MHz band                           |
| DCT      | Discrete Cosine Transform                                              |
| DDF      | Diversity-Decode-and-Forward                                           |
| DDIR     | Double Directional Impulse Response                                    |
| DDDPS    | Double Directional Delay Power Spectrum                                |
| DECT     | Digital Enhanced Cordless Telecommunications (ETSI)                    |
| DF       | Decode-and-Forward                                                     |
| DFE      | Decision Feedback Equalizer                                            |
| DFT      | Discrete Fourier Transform                                             |
| DIFS     | Distributed Inter Frame Space                                          |
| DL       | Downlink                                                               |
| DLL      | Data Link Layer                                                        |
| DLT      | Digital Ledger Technology                                              |
| DM       | Delta Modulation                                                       |
| DMC      | Diffuse Multipath Component (Chapters 6–9)                             |
| DMC      | Discrete Memoryless Channel (Chapters 10–13)                           |

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|         |                                                                     |
|---------|---------------------------------------------------------------------|
| DMT     | Discrete Multi Tone                                                 |
| DNS     | Domain Name Server                                                  |
| DOA     | Direction Of Arrival                                                |
| DOD     | Direction Of Departure                                              |
| DODAG   | Destination-Oriented Directed Acyclic Graph                         |
| DPC     | Dirty Paper Coding                                                  |
| DPSK    | Differential Phase Shift Keying                                     |
| DQPSK   | Differential Quadrature-Phase Shift Keying                          |
| DRM     | Discontinuous Reception Mechanisms                                  |
| DRT     | Diagnostic Rhyme Test                                               |
| DRX     | Discontinuous Reception                                             |
| DS      | Direct Sequence                                                     |
| DS-CDMA | Direct Sequence-Code Division Multiple Access                       |
| DS-SS   | Direct Sequence-Spread Spectrum                                     |
| DSA     | Dynamic Spectrum Access                                             |
| DSDV    | Destination-Sequenced Distance Vector                               |
| DSI     | Digital Speech Interpolation                                        |
| DSL     | Digital Subscriber Line                                             |
| DSMA    | Data Sense Multiple Access                                          |
| DSP     | Digital Signal Processor                                            |
| DSR     | Distributed Speech Recognition (Chapter 24)                         |
| DSR     | Dynamic Source Routing (Chapter 23)                                 |
| DTAP    | Direct Transfer Application Part                                    |
| DTB     | Dynamic Tuning Backoff                                              |
| DTE     | Data Terminal Equipment                                             |
| DtFT    | Discrete-time Fourier Transform                                     |
| DTMF    | Dual Tone Multi Frequency (signalling)                              |
| DTX     | Discontinuous Transmission                                          |
| DUT     | Device Under Test                                                   |
| DVB     | Digital Video Broadcasting                                          |
| DVB-H   | Digital Video Broadcasting – Handheld                               |
| DVD     | Digital Video Disk                                                  |
| DxF     | Diversity xF                                                        |
| E-CID   | Enhanced Cell Identification                                        |
| EC      | European Commission                                                 |
| ECL     | Emitter Coupled Logic                                               |
| EDCA    | Enhanced Distributed Channel Access                                 |
| EDGE    | Enhanced Data rates for GSM Evolution                               |
| EFFR    | Enhanced Fractional Frequency Reuse                                 |
| EFIM    | Equivalent Fisher Information Matrix                                |
| EFR     | Enhanced Full Rate                                                  |
| EGC     | Equal Gain Combining                                                |
| EIA     | Electronic Industries Alliance (U.S.A.)                             |
| EIR     | Equipment Identity Register                                         |
| EIRP    | Equivalent Isotropically Radiated Power                             |
| EKF     | Extended Kalman Filter                                              |
| ELP     | Equivalent Low Pass                                                 |
| EM      | Expectation-Maximization (Chapter 9)                                |
| EM      | ElectroMagnetic                                                     |
| EMS     | Enhanced Messaging Service                                          |
| EN      | European Norm                                                       |
| ENOB    | Effective Number of Bits                                            |
| EPC     | Electronic Product Code                                             |
| ERLE    | Echo Return Loss Enhancement                                        |
| ERP     | EquiRectangle Projection                                            |
| ESN     | Electronic Serial Number                                            |
| ESPRIT  | Estimation of Signal Parameters by Rotational Invariance Techniques |
| ETS     | European Telecommunication Standard                                 |
| ETSI    | European Telecommunications Standards Institute                     |
| ETX     | Expected Number of Transmissions                                    |
| EV-DO   | EVolution-Data Optimized                                            |

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|          |                                                              |
|----------|--------------------------------------------------------------|
| EVD      | Eigen Value Decomposition                                    |
| EVM      | Error Vector Magnitude                                       |
| EVRC     | Enhanced Variable Rate Coder                                 |
| F0       | Fundamental frequency                                        |
| FB       | Frequency correction Burst                                   |
| FBMC     | Filter Bank Multicarrier                                     |
| FCC      | Federal Communications Commission                            |
| FD-MIMO  | Full-Dimensional MIMO                                        |
| FDD      | Frequency Domain Duplexing                                   |
| FDMA     | Frequency Division Multiple Access                           |
| FDTD     | Finite Difference Time Domain                                |
| FEC      | Forward Error Correction                                     |
| FEM      | Finite Element Method                                        |
| FFR      | Fractional Frequency Reuse                                   |
| FFR-FI   | Fractional Frequency Reuse Full Isolation                    |
| FFT      | Fast Fourier Transform                                       |
| FH       | Frequency Hopping                                            |
| FHMA     | Frequency Hopping Multiple Access                            |
| FI       | Floating Intercept                                           |
| FIFO     | First In First Out                                           |
| FIM      | Fisher Information Matrix                                    |
| FIR      | Finite Impulse Response                                      |
| FL       | Federated Learning                                           |
| FM       | Frequency Modulation                                         |
| FMT      | Filtered MultiTone                                           |
| FR       | Full Rate                                                    |
| FS       | Federal Standard                                             |
| FSK      | Frequency Shift Keying                                       |
| FT       | Fourier Transform                                            |
| FTF      | Fast Transversal Filter                                      |
| FWA      | Fixed Wireless Access                                        |
| FWHM     | Full Width Half Maximum                                      |
| GDOP     | Geometric Dilution Of Precision                              |
| GF       | Galois Field                                                 |
| GFDM     | Generalized Frequency Division Multiplexing                  |
| GFSK     | Gaussian Frequency Shift Keying                              |
| GLONASS  | Global Navigation Satellite System                           |
| GLP      | Grassmannian Line Packing                                    |
| GNSS     | Global Navigation Satellite System                           |
| GP       | Global Parameter                                             |
| GPRS     | General Packet Radio Service                                 |
| GPS      | Global Positioning System                                    |
| GPU      | Graphic Processing Unit                                      |
| GRU      | Gated Recurrent Unit                                         |
| GSC      | Generalized Selection Combining                              |
| GSCM     | Geometry-based Stochastic Channel Model                      |
| GSM      | Global System for Mobile communications                      |
| GSM PLMN | GSM Public Land Mobile Network                               |
| GSM 1800 | Global System for Mobile communications at the 1800-MHz band |
| GTP      | GPRS Tunneling Protocol                                      |
| H-BLAST  | Horizontal BLAST                                             |
| H-S/MRC  | Hybrid Selection/Maximum Ratio Combining                     |
| HARQ     | Hybrid Automatic Repeat reQuest                              |
| HD       | High Definition                                              |
| HDR      | High Dynamic Range                                           |
| HDTV     | High Definition TeleVision                                   |
| HF       | High Frequency                                               |
| HIPERLAN | High PERformance Local Area Network                          |
| HNМ      | Harmonic + Noise Modeling                                    |
| hostid   | host address                                                 |
| HO       | HandOver                                                     |

|              |                                                                         |
|--------------|-------------------------------------------------------------------------|
| HPA          | High Power Amplifier                                                    |
| HPPP         | Homogeneous Poisson Point Process                                       |
| HRPE         | High Resolution Parameter Estimation                                    |
| HRTF         | Head Related Transfer Function                                          |
| HSDPA        | High Speed Downlink Packet Access                                       |
| HSPA         | High-Speed Packet Access                                                |
| HT           | High Throughput                                                         |
| HT           | Hilly Terrain (Chapter 7)                                               |
| HTTP         | Hyper Text Transfer Protocol                                            |
| IAF          | InterSymbol Interference Amplify-and-Forward                            |
| IAM          | Initial Address Message                                                 |
| IC           | Integrated Circuit                                                      |
| ICI          | Inter Carrier Interference (Chapter 15)                                 |
| ICI          | Inter Cell Interference (Chapter 22)                                    |
| ICIC         | Inter Cell Interference Coordination                                    |
| ICN          | Information-Centric and Intent-Based Networks                           |
| ID           | Identification/Identifier                                               |
| IDFT         | Inverse Discrete Fourier Transform                                      |
| IE           | Information Element                                                     |
| IEC          | International Electrotechnical Commission                               |
| IEEE         | Institute of Electrical and Electronics Engineers                       |
| IETF         | Internet Engineering Task Force                                         |
| IF           | Intermediate Frequency                                                  |
| IFF          | Identification Friend or Foe                                            |
| IFFT         | Inverse Fast Fourier Transformation                                     |
| IFR          | Incremental Frequency Reuse                                             |
| iid          | independent identically distributed                                     |
| IIR          | Infinite Impulse Response                                               |
| ILBC         | Internet Low Bit-rate Codec                                             |
| IMBE         | Improved Multi Band Excitation                                          |
| IMEI         | International Mobile station Equipment Identity                         |
| IMSI         | International Mobile Subscriber Identity                                |
| IMT          | International Mobile Telecommunications                                 |
| IMT-2000     | International Mobile Telecommunications 2000 (standard)                 |
| IMT-2020     | International Mobile Telecommunications 2020 (standard)                 |
| IMT-Advanced | International Mobile Telecommunications Advanced (standard)             |
| IMU          | Inertial Measurement Unit                                               |
| INMARSAT     | INternational MARitime SATellite System                                 |
| INR          | Interference to Noise Ratio                                             |
| IO           | Interacting Object                                                      |
| I/O          | Input/Output                                                            |
| IoT          | Internet of Things                                                      |
| IP           | Internet Protocol                                                       |
| IP           | Intercept Point (Chapter 17)                                            |
| IPO          | Initial Public Offering                                                 |
| IPR          | Intellectual Property Rights                                            |
| IQ           | In-Phase – Quadrature Phase                                             |
| IR           | Impulse Radio                                                           |
| IRIDIUM      | an early satellite communication system                                 |
| IRS          | Intelligent Reflective Surface                                          |
| IS-95        | Interim Standard 95 (the first CDMA system adopted by the American TIA) |
| ISDN         | Integrated Services Digital Network                                     |
| ISI          | InterSymbol Interference                                                |
| ISM          | Industrial, Scientific, and Medical                                     |
| ISO          | International Standards Organization                                    |
| ISPP         | Interleaved Single Pulse Permutation                                    |
| ITLinQ       | a scheduling algorithm for ad-hoc networks                              |
| ITU          | International Telecommunications Union                                  |
| ITU-R        | ITU – Radio                                                             |
| ITU-T        | ITU – Telecommunications                                                |
| IWF          | Inter Working Function                                                  |

|                |                                                             |
|----------------|-------------------------------------------------------------|
| IWU            | Inter Working Unit                                          |
| IxF            | Interference xF                                             |
| JD-TCDMA       | Joint Detection–Time/Code Division Multiple Access          |
| JDC            | Japanese Digital Cellular                                   |
| JPEG           | Joint Photographic Expert Group                             |
| JSDM           | Joint Spatial Division and Multiplexing                     |
| JVT            | Joint Video Team                                            |
| KPI            | Key Performance Indicator                                   |
| KS             | Kolmogorov–Smirnov                                          |
| KLT            | Karhunen Loève Transform                                    |
| LAN            | Local Area Network                                          |
| LAR            | Logarithmic Area Ratio                                      |
| LASSO          | Least Absolute Shrinkage and Selection Operator             |
| LATS           | Location Aware Training Scheme                              |
| LBG            | Linde–Buzo–Gray algorithm                                   |
| LBT            | Listen Before Talk                                          |
| LCR            | Level Crossing Rate                                         |
| LD-CELP        | Low Delay–Code Excited Linear Prediction                    |
| LDPC           | Low Density Parity Check                                    |
| LDS            | Low Density Spreading                                       |
| LEACH          | Low-Energy Adaptive Clustering Hierarchy (routing protocol) |
| LED            | Light Emitting Diode                                        |
| LEO            | Low Earth Orbit                                             |
| LFSR           | Linear Feedback Shift Register                              |
| LIFO           | Last In First Out                                           |
| LILD           | Linear Increase Linear Decrease                             |
| LLC            | Logical Link Control                                        |
| LLR            | Log Likelihood Ratio                                        |
| LMMSE          | Linear Minimum Mean Square Error                            |
| LMS            | Least Mean Square                                           |
| LNA            | Low Noise Amplifier                                         |
| LO             | Local Oscillator                                            |
| LPF            | LowPass Filter                                              |
| LoRa           | Long Range (standard for IoT)                               |
| LOS            | Line Of Sight                                               |
| LP             | Local Parameter (Chapter 7)                                 |
| LP             | Linear Prediction/Predictor (Chapter 24)                    |
| LP             | Linear Program (Chapter 23)                                 |
| LP (subscript) | Low-pass                                                    |
| LPC            | Linear Predictive voCoder                                   |
| LR             | Location Register                                           |
| LS             | Least Squares                                               |
| LSB            | Least Significant Bit                                       |
| LSF            | Line Spectral Frequency                                     |
| LSP            | Large-Scale Parameters                                      |
| LSP            | Line Spectrum Pair (Chapter 24)                             |
| LSTM           | Long Short Term Memory                                      |
| LTE            | Long-Term Evolution                                         |
| LTI            | Linear Time Invariant                                       |
| LTP            | Long Term Prediction/Predictor                              |
| LTV            | Linear Time Variant                                         |
| M-QAM          | M-ary Quadrature Amplitude Modulation                       |
| MA             | Multiple Access                                             |
| MAC            | Medium Access Control                                       |
| MAF            | Multihop Amplify-and-Forward                                |
| MAHO           | Mobile Assisted Hand Over                                   |
| MAI            | Multiple Access Interference                                |
| MAN            | Metropolitan Area Network                                   |
| MANET          | Mobile Ad-hoc NETwork                                       |
| MAP            | Maximum A Posteriori                                        |
| MB             | MacroBlock                                                  |

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|         |                                                                                       |
|---------|---------------------------------------------------------------------------------------|
| MBE     | Multi Band Excitation                                                                 |
| MBOA    | Multi Band OFDM Alliance                                                              |
| MC      | Mobile Client                                                                         |
| MC-CDMA | Multi Carrier Code Division Multiple Access                                           |
| MCS     | Modulation and Coding Scheme                                                          |
| MDC     | Multiple Description Coding                                                           |
| MDF     | Multihop Decode-and-Forward                                                           |
| MDHO    | Macro Diversity HandOver                                                              |
| MDM     | Mode Division Multiplexing                                                            |
| MDS     | Minimum Detectable Signal                                                             |
| MEA     | Multiple Element Antenna                                                              |
| MEC     | Mobile Edge Computing                                                                 |
| MEF     | Maintenance Entity Function                                                           |
| MEG     | Mean Effective Gain                                                                   |
| MELP    | Mixed Excitation Linear Prediction                                                    |
| MEMS    | Micro Electro-Mechanical Switch                                                       |
| METIS   | Mobile and wireless communications Enablers for the Twenty–twenty Information Society |
| MFEP    | Matched Front End Processor                                                           |
| MG      | Mesh Gateway                                                                          |
| MGF     | Moment Generating Function                                                            |
| MILD    | Multiplicative Increase Linear Decrease                                               |
| MIME    | Multipurpose Internet Mail Extensions                                                 |
| MIMO    | Multiple Input Multiple Output system                                                 |
| MIPS    | Million Instructions Per Second                                                       |
| MISO    | Multiple Input Single Output                                                          |
| MiWEBA  | Millimetre-wave Evolution for Backhaul and Access                                     |
| ML      | Maximum Likelihood (Chapters 11, 14, 16, 29.1–5)                                      |
| ML      | Machine Learning (Chapters 4, 20, 26, 29.11, 35)                                      |
| MLE     | Maximum Likelihood Estimation                                                         |
| MLP     | Multilayer Perceptron                                                                 |
| MLSE    | Maximum Likelihood Sequence Estimators (or Estimation)                                |
| MMS     | Multimedia Messaging Service                                                          |
| MMSE    | Minimum Mean Square Error                                                             |
| MNRU    | Modulated Noise Reference Unit                                                        |
| MOS     | Mean Opinion Score                                                                    |
| MoU     | Memorandum of Understanding                                                           |
| MP3     | Motion Picture Experts Group-1 Layer 3                                                |
| MPC     | Multi Path Component                                                                  |
| MPD     | Media Presentation Description                                                        |
| MPDU    | MAC Protocol Data Unit                                                                |
| MPEG    | Motion Picture Experts Group                                                          |
| MPR     | Multipoint Relay                                                                      |
| MPSK    | M-ary Phase Shift Keying                                                              |
| MR      | Mesh Router                                                                           |
| MRC     | Maximum Ratio Combining                                                               |
| MRT     | Maximum Ratio Transmission                                                            |
| MS      | Mobile Station (synonymous for UE)                                                    |
| MS ISDN | Mobile Station ISDN Number                                                            |
| MSB     | Most Significant Bit                                                                  |
| MSC     | Mobile Switching Center                                                               |
| MSDU    | MAC Service Data Unit                                                                 |
| MSE     | Mean Square Error                                                                     |
| MSIN    | Mobile Subscriber Identification Number                                               |
| MSISDN  | Mobile Station ISDN Number                                                            |
| MSK     | Minimum Shift Keying                                                                  |
| MSS     | Mobile Satellite Service                                                              |
| MU-MIMO | Multiuser Multiple-Input Multiple-Output                                              |
| MUMS    | Multiuser Mobile Station                                                              |
| MUSA    | Multiuser Shared Access                                                               |
| MUSIC   | Multiple Signal Classification                                                        |
| MUX     | Multiplexing                                                                          |



|         |                                               |
|---------|-----------------------------------------------|
| MVC     | Multiview Video Coding                        |
| MVM     | Minimum Variance Method                       |
| MxF     | Multihop xF                                   |
| NAL     | Network Abstraction Layer                     |
| NB      | Narrow Band                                   |
| NDxF    | Nonorthogonal Diversity xF                    |
| netid   | network address                               |
| NF      | Network Function                              |
| NFC     | Near Field Communications                     |
| NFV     | Network Function Virtualization               |
| NLOS    | Non Line Of Sight                             |
| NLP     | Non Linear Processor                          |
| NM      | Network Management                            |
| NMSE    | Normalized Mean Square Error                  |
| NMT     | Nordic Mobile Telephone                       |
| NN      | Neural Network                                |
| NOMA    | Non-Orthogonal Multiple Access                |
| NP      | Non-deterministic Polynomial Time             |
| Node-B  | base station                                  |
| NR      | New Radio (the 5G standard of 3GPP)           |
| NRZ     | Non Return to Zero                            |
| NS-OFDM | Null Suffix OFDM                              |
| NTT     | Nippon Telephone and Telegraph                |
| O&M     | Operations and Maintenance                    |
| O2I     | Outdoor-to-Indoor                             |
| OAM     | Orbital Angular Momentum                      |
| OEM     | Original Equipment Manufacturer               |
| OFDM    | Orthogonal Frequency Division Multiplexing    |
| OFDMA   | Orthogonal Frequency Division Multiple Access |
| OLOS    | Obstructed Line of Sight                      |
| OLSR    | Optimized Link State Routing                  |
| OMA     | Orthogonal Multiple Access                    |
| OOK     | On Off Keying                                 |
| OPT     | Operator Perturbation Technique               |
| OQAM    | Offset Quadrature Amplitude Modulation        |
| OQPSK   | Offset Quadrature Phase Shift Keying          |
| OS      | Operating Systems                             |
| OSI     | Operator System Interface                     |
| OTD     | Orthogonal Transmit Diversity                 |
| OTDOA   | Observed Time Difference Of Arrival           |
| OTFS    | Orthogonal Time Frequency Space               |
| OVSF    | Orthogonal Variable Spreading Factor          |
| P/S     | Parallel to Serial (conversion)               |
| PA      | Power Amplifier                               |
| PABX    | Private Automatic Branch eXchange             |
| PAM     | Pulse Amplitude Modulation                    |
| PAN     | Personal Area Network                         |
| PAPR    | Peak-to-Average Power Ratio                   |
| PAR     | Peak-to-Average Ratio                         |
| PAR     | Project Authorization Request (Chapter 30)    |
| PARC    | Per Antenna Rate Control                      |
| PARCOR  | PARTial CORrelation                           |
| PBPC    | Per Base station Power Constraint             |
| PCB     | Printed Circuit Board                         |
| PCM     | Pulse Code Modulated                          |
| PCS     | Personal Communication System                 |
| PDA     | Personal Digital Assistant                    |
| PDMA    | Pattern Division Multiple Access              |
| PDC     | Pacific Digital Cellular (Japanese system)    |
| pdf     | probability density function                  |
| PDP     | Power Delay Profile                           |

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|        |                                                   |
|--------|---------------------------------------------------|
| PDU    | Protocol Data Unit                                |
| PESQ   | Perceptual Evaluation of Speech Quality           |
| PF     | Proportional Fair                                 |
| PFR    | Partial Frequency Reuse                           |
| PGF    | Probability Generating Functional                 |
| PHS    | Personal Handyphone System                        |
| PHY    | PHYSical layer                                    |
| PIC    | Parallel Interference Cancellation                |
| PIFA   | Planar Inverted F Antenna                         |
| PIN    | Personal Identification Number                    |
| PLCP   | Physical Layer Convergence Procedure              |
| PLL    | Phase Locked Loop                                 |
| PLMN   | Public Land Mobile Network                        |
| PLS    | Physical Layer Security                           |
| PM     | Phase Modulation                                  |
| PN     | Pseudo Noise                                      |
| POCS   | Projection Onto Convex Sets                       |
| POP    | Peak to Off Peak                                  |
| POTS   | Plain Old Telephone System                        |
| PPDU   | Physical layer Protocol Data Unit                 |
| PPM    | Pulse Position Modulation                         |
| PPP    | Poisson Point Process                             |
| PRake  | Partial Rake                                      |
| PRMA   | Packet Reservation Multiple Access                |
| PRS    | Position Reference Signal                         |
| PSD    | Power Spectral Density                            |
| PSDU   | Physical Layer Service Data Unit                  |
| PSK    | Phase Shift Keying                                |
| PSPDN  | Packet Switched Public Data Network               |
| PSQM   | Perceptual Speech Quality Measurement             |
| PSTN   | Public Switched Telephone Network                 |
| PSWF   | Prolate Spheriodal Wave Function                  |
| PTM    | Point To Multipoint                               |
| PTM-M  | Point To Multipoint Multicast                     |
| PTM-SC | Point To Multipoint Service Center                |
| PTO    | Public Telecommunications Operators               |
| PUK    | Personal Unblocking Key                           |
| PWI    | Prototype Waveform Interpolation                  |
| PWT    | Personal Wireless Telephony (standardized system) |
| QAM    | Quadrature Amplitude Modulation                   |
| QAP    | QoS Access Point                                  |
| QCELP  | Qualcomm Code Excited Linear Prediction           |
| QFGV   | Quadratic Form Gaussian Variable                  |
| QoE    | Quality of Experience                             |
| QOF    | Quasi Orthogonal Function                         |
| QoS    | Quality of Service                                |
| QPSK   | Quadrature-Phase Shift Keying                     |
| QSNR   | Quantization SNR                                  |
| RA     | Rural Area (Chapter 7)                            |
| RA     | Random Access                                     |
| RAN    | Radio Access Network                              |
| RAND   | Reasonable And Non-Discriminatory                 |
| RBF    | Radial Basis Function                             |
| RC     | Raised Cosine                                     |
| RCDLA  | Radiation Coupled Dual L Antenna                  |
| RE     | Radio Environment (Chapters 5–9)                  |
| RE     | Resource Element (Chapters 30–35)                 |
| RF     | Radio Frequency                                   |
| RFID   | Radio Frequency Identification                    |
| RFL    | Radio Frequency subLayer                          |
| RFN    | Reduced TDMA Frame Number                         |

|         |                                                          |
|---------|----------------------------------------------------------|
| RGB     | Red Green Blue                                           |
| RiMax   | a maximum-likelihood parameter estimation algorithm      |
| RL      | Reinforcement Learning                                   |
| RLC     | Radio Link Control                                       |
| RLP     | Radio Link Protocol                                      |
| RLS     | Recursive Least Squares                                  |
| RMSE    | Root Mean Square Error                                   |
| RNN     | Recurrent Neural Network                                 |
| ROM     | Read Only Memory                                         |
| RPAR    | Relay PAtH Routing                                       |
| RPE     | Regular Pulse Excitation (voice codec)                   |
| RPE-LTP | Regular Pulse Excited with Long Term Prediction          |
| RPL     | Routing Protocol for Low-Power and Lossy Networks        |
| RR      | Round Robin                                              |
| RRC     | Radio Resource Control                                   |
| RS      | Reed–Solomon (code) (Chapters 13, 25, 34)                |
| RS      | Reference Signal (Chapters 31, 32)                       |
| RS      | Relay Station (Chapter 27)                               |
| RS      | Rate Splitting (Chapter 35)                              |
| RSC     | Recursive Systematic Convolutional (Chapter 13)          |
| RSC     | Radio Spectrum Committee (Chapter 26)                    |
| RSSI    | Received Signal Strength Indication                      |
| RTP     | Real-time Transport Protocol                             |
| RTCP    | Control part of RTP                                      |
| RTS     | Request to Send                                          |
| RTSP    | Real Time Streaming Protocol                             |
| RTP     | Real-time Transport Protocol                             |
| rv      | random variable                                          |
| RVLC    | Reversible Variable Length Code                          |
| RX      | Receiver                                                 |
| RXLEV   | Received signal LEVel                                    |
| RXQUAL  | Received signal QUALity                                  |
| SA      | Service and system Aspects                               |
| SAGE    | Space Alternating Generalized Expectation – maximization |
| SAP     | Service Access Point                                     |
| SAR     | Specific Absorption Rate                                 |
| SASB    | Standards Association Standards Board (of the IEEE)      |
| SAW     | Surface Acoustic Wave                                    |
| SC-CDMA | Single-Carrier CDMA                                      |
| SC-FDMA | Single-Carrier FDMA                                      |
| SCD     | Successive Cancellation Decoding                         |
| SCMA    | Sparse Code Multiple Access                              |
| SCxF    | Split-Combine xF                                         |
| SDMA    | Space Division Multiple Access                           |
| SDN     | Software Defined Networking                              |
| SDR     | Software Defined Radio                                   |
| SDU     | Service Data Unit                                        |
| SEI     | Supplemental Enhancement Information                     |
| SEP     | Symbol Error Probability                                 |
| SER     | Symbol Error Rate                                        |
| SF      | Spreading Factor                                         |
| SFBC    | Space Frequency Block Coding                             |
| SFFT    | Symplectic Fast Fourier Transform                        |
| SFIR    | Spatial Filtering for Interference Reduction             |
| SFR     | Soft Frequency Reuse                                     |
| SGD     | Stochastic Gradient Descent                              |
| SIC     | Successive (or Serial) Interference Cancellation         |
| SID     | Silence Descriptor                                       |
| SIG     | Special Interest Group                                   |
| SIM     | Subscriber Identity Module                               |
| SIMO    | Single Input Multiple Output                             |

|          |                                                         |
|----------|---------------------------------------------------------|
| SINR     | Signal-to-Interference-and-Noise Ratio                  |
| SIR      | Signal-to-Interference Ratio                            |
| SISO     | Soft Input Soft Output (Chapter 13)                     |
| SISO     | Single Input Single Output                              |
| SLAM     | Simultaneous Location and Mapping                       |
| SLNR     | Signal-to-Leakage and Noise Ratio                       |
| SM       | Spatial Multiplexing                                    |
| SMG      | Spatial Multiplexing Gain                               |
| SMS      | Short Message Service                                   |
| SMTp     | Short Message Transfer Protocol                         |
| SN       | Serial Number                                           |
| SNR      | Signal-to-Noise Ratio                                   |
| SOLT     | Short Open Loss Termination                             |
| SON      | Self Organizing Network                                 |
| SP       | Shortest Path                                           |
| SPC      | Sum Power Constraint                                    |
| SPEB     | Square Position Error Bound                             |
| SPP      | Spherical Phase Plate                                   |
| S/P      | Serial to Parallel (conversion)                         |
| SQNR     | Signal-to-Quantization Noise Ratio                      |
| SR       | Spatial Reference                                       |
| SRake    | Selective Rake                                          |
| SRMA     | Split-channel Reservation Multiple Access               |
| SRS      | Sounding Reference Signal                               |
| SSA      | Small Scale Averaged                                    |
| SSF      | Small-Scale Fading                                      |
| ST       | Space – Time                                            |
| STA      | STation                                                 |
| STBC     | Space Time Block Code                                   |
| STC      | Sinusoidal Transform Coder                              |
| STDCC    | Swept Time Delay Cross Correlator                       |
| STP      | Short Term Prediction/Predictor                         |
| STS      | Space Time Spreading                                    |
| STTC     | Space Time Trellis Code                                 |
| SU-MIMO  | Single-User MIMO                                        |
| SV       | Saleh-Valenzuela model                                  |
| SVD      | Singular Value Decomposition                            |
| SVM      | Support Vector Machine                                  |
| SWIPT    | Simultaneous Wireless Information and Power Transfer    |
| TC       | Topology Control                                        |
| TCM      | Trellis Coded Modulation                                |
| TCP      | Transmission Control Protocol                           |
| TD-SCDMA | Time Division-Synchronous Code Division Multiple Access |
| TDD      | Time Domain Duplexing                                   |
| TDMA     | Time Division Multiple Access                           |
| TDOA     | Time Difference Of Arrival                              |
| TE       | Transversal Electric                                    |
| TETRA    | TErrestrial TRunked RAdio                               |
| TF       | Time Frequency                                          |
| TH       | Tomlinson-Harashima                                     |
| TH-IR    | Time Hopping Impulse Radio                              |
| TIA      | Telecommunications Industry Association (U.S.)          |
| TL       | Transfer Learning                                       |
| TM       | Transversal Magnetic                                    |
| TMSI     | Temporary Mobile Subscriber Identity                    |
| TOA      | Time Of Arrival                                         |
| TPC      | Transmit Power Control                                  |
| TR       | Technical Report (ETSI)                                 |
| TR       | Temporal Reference (Chapter 16)                         |
| TR       | Transmitted Reference (Chapter 19)                      |
| TS       | Time Slot                                               |

|         |                                                    |
|---------|----------------------------------------------------|
| TTA     | Telecommunications Technology Association of Korea |
| TTC     | Telecommunications Technology Committee            |
| TTS     | Text To Speech Synthesis                           |
| TU      | Typical Urban                                      |
| TWDP    | Two-path With Diffuse Power                        |
| TX      | Transmitter                                        |
| TXOP    | Transmission OPportunity                           |
| U-NII   | Unlicensed National Information Infrastructure     |
| UAV     | Unmanned Arial Vehicle                             |
| UDP     | User Datagram Protocol                             |
| UE      | User Equipment                                     |
| UE-ID   | User Equipment in-band IDentification              |
| UFMC    | Universal Filtered MultiCarrier                    |
| UL      | Uplink                                             |
| ULA     | Uniform Linear Array                               |
| UMB     | Ultra Mobile Broadband (standard)                  |
| UMTS    | Universal Mobile Telecommunications System         |
| US      | Uncorrelated Scatterer                             |
| USB     | Universal Serial Bus                               |
| UTDOA   | Uplink Time Difference Of Arrival                  |
| UTRA    | UMTS Terrestrial Radio Access                      |
| UTRAN   | UMTS Terrestrial Radio Access Network              |
| UWB     | Ultra Wide Bandwidth                               |
| VA      | Virtual Anchor                                     |
| VAD     | Voice Activity Detection/Detector                  |
| VCDA    | Virtual Cell Deployment Area                       |
| VCEG    | Video Coding Expert Group                          |
| VCO     | Voltage Controlled Oscillator                      |
| VGA     | Variable Gain Amplifier                            |
| VL      | Very Long (to indicate type of Kasami sequences)   |
| VLC     | Variable Length Coding                             |
| VLC     | Visible Light Communications (Chapters 30, 35)     |
| VNA     | Vector Network Analyzer                            |
| VoIP    | Voice over Internet Protocol                       |
| VR      | Virtual Reality                                    |
| VR      | Visibility Region (Chapter 7)                      |
| VRB     | Virtual Resource Block                             |
| VQ      | Vector Quantization/Quantizer                      |
| VSELP   | Vector Sum Excited Linear Prediction               |
| VSWR    | Voltage Standing Wave Ratio                        |
| VVC     | Versatile Video Coding                             |
| WB      | Wide Band                                          |
| WCDMA   | Wideband Code Division Multiple Access             |
| WF      | Whitening Filter                                   |
| WG      | Working Group                                      |
| WH      | Walsh-Hadamard                                     |
| WI      | Waveform Interpolation                             |
| Wi-Fi   | Wireless Fidelity                                  |
| WLAN    | Wireless Local Area Network                        |
| WLL     | Wireless Local Loop                                |
| WM      | Wireless Medium                                    |
| WMMSE   | Weighted MMSE                                      |
| WRC     | World Radio Conference                             |
| WSS     | Wide Sense Stationary                              |
| WSSUS   | Wide Sense Stationary Uncorrelated Scatterer       |
| xAI     | explainable Artificial Intelligence                |
| XPd     | Cross-Polarization Discrimination                  |
| YCgCo-R | A color space                                      |
| YIG     | Yttrium Iron Garnett                               |
| ZC      | Zadoff-Chu                                         |
| ZF      | Zero-Forcing                                       |